Ex. No: 7 Banker's Algorithm

C PROGRAM:

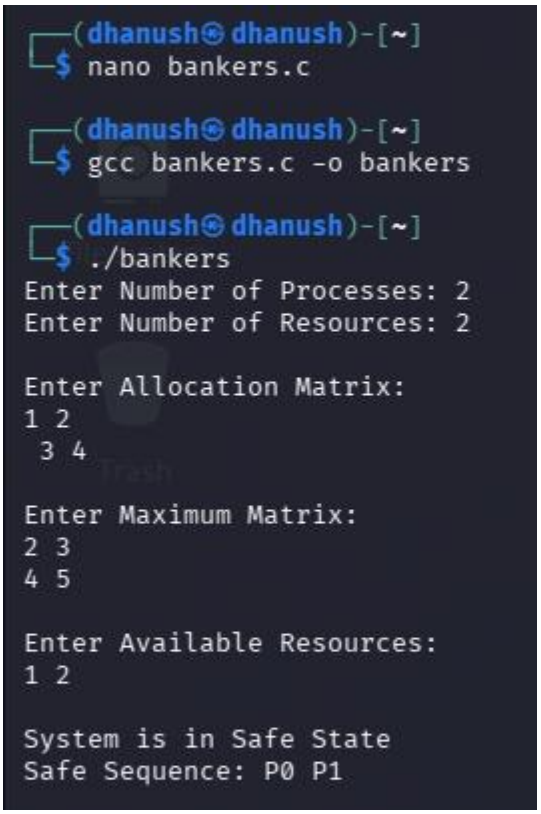
```
#include <stdio.h>
int main()
{
int n, m, i, j, k;
int allocation[10][10], max[10][10];
int need[10][10];
int available[10];
int finish[10] = {0};
int safeSeq[10];
int count = 0;
printf("Enter Number of Processes: ");
scanf("%d", &n);
printf("Enter Number of Resources: ");
scanf("%d", &m);
printf("
\nEnter Allocation Matrix:
\n");

for(i = 0; i < n; i++)
for(j = 0; j < m; j++)
scanf("%d", &allocation[i][j]);
printf("
```

```
\nEnter Maximum Matrix:
\n");

for(i = 0; i < n; i++)
for(j = 0; j < m; j++)
scanf("%d", &max[i][j]);
printf("
\nEnter Available Resources:
\n");

for(i = 0; i < m; i++)
scanf("%d", &available[i]);
for(i = 0; i < n; i++)
{
for(j = 0; j < m; j++)
{
need[i][j] = max[i][j]

- allocation[i][j];

}
}
while(count < n)
{
int found = 0;
for(i = 0; i < n; i++)
```

```
{
if(finish[i] == 0)
{
for(j = 0; j < m; j++)
{
if(need[i][j] > available[j])
break;
}
if(j == m)
{
for(k = 0; k < m; k++)
available[k] += allocation[i][k];
safeSeq[count++] = i;
finish[i] = 1;
found = 1;
}
}
}
if(found == 0)
{
printf("\nSystem is NOT in Safe State\n");
return 0;
}
}
printf("\nSystem is in Safe State\n");
printf("Safe Sequence: ");
```

```
for(i = 0; i < n; i++)

printf("P%d ", safeSeq[i]);

printf("\n");

return 0;

}
```

Output:

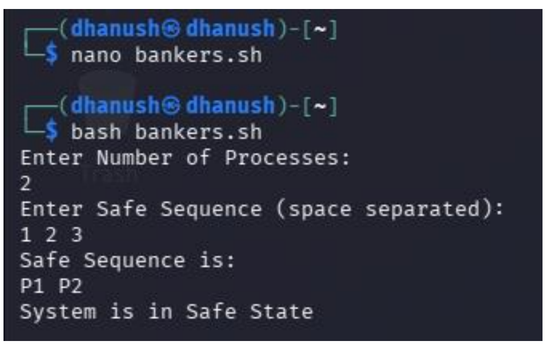


SHELL SCRIPT

```
#!/bin/bash

echo "Enter Number of Processes:"
```

```
read n

echo "Enter Safe Sequence (space separated):"

read -a seq

echo "Safe Sequence is:"

for ((i=0;i<n;i++))

do

echo -n "P${seq[$i]} "

done

echo

echo "System is in Safe State"
```

Output:



```
┌──(dhanush㉿dhanush)-[~]
└─$ nano bankers.sh

┌──(dhanush㉿dhanush)-[~]
└─$ bash bankers.sh
Enter Number of Processes:
2
Enter Safe Sequence (space separated):
1 2 3
Safe Sequence is:
P1 P2
System is in Safe State
```